YQuantum 2026 QuEra Challenge Report: One Qubit, Many Problems

Marlon Jost[1, 2], Austin Jackson[1], Claudia Friedsam[3], Noble Agyeman-Bobie[4], and Marcos Velazquez Ramos[5]

[1]Khoury College of Computer Sciences, Northeastern University, Boston, Massachusetts 02115, United States
[2]Department of Physics, Northeastern University, Boston, Massachusetts 02115, United States
[3]Independent Researcher, Newton, MA 02461
[4]Department of Mathematics and Physics, Grambling State University, Grambling, Louisiana, 71245, United States
[5]School of Computer, Data, Information Sciences, University of Wisconsin Madison, Madison, Wisconsin, 53706

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1 Introduction

When Hamlet and Qoratio encountered the ghost of the old physicist in Yale's courtyards, they heard a claim that seemed almost too simple to be interesting:

> *"A 1-qubit gate can be harder to implement than a 2-qubit gate."*

Their first instinct was skepticism, and rightly so. A 1-qubit gate acts on a single qubit; a 2-qubit gate must coordinate two. A 1-qubit gate is a 2×2 unitary matrix; a 2-qubit gate is 4×4. By almost any naive measure of complexity, the 1-qubit gate should be easier. Yet the ghost's parting whisper—*"think logical qubits"*—contained the key: the difficulty is not in the gate itself, but in the context in which it must be applied.
This write-up traces the full arc of that difficulty. We focus on the family of Z-rotations

$$R_z(\pi/2^k), \qquad k \in \{0, 1, 2, 3, 4, 5\}, \tag{1}$$

and follow the challenge they pose across three compounding layers of hardness.

Layer 1: Synthesis cost. On a physical qubit, $R_z(\pi/2^k)$ is not always available as a native gate. The standard universal gate set for fault-tolerant quantum computing is Clifford+T := $\{H, S, T, \text{CNOT}\}$ [1]. Within this gate set, the Clifford gates $\{H, S, \text{CNOT}\}$ are "cheap"—they can be implemented fault-tolerantly with modest overhead—while the T gate carries a disproportionate cost. Approximating $R_z(\pi/2^k)$ to precision ε requires $\Theta(\log(1/\varepsilon))$ T gates [6], a cost that grows without bound as the target angle shrinks or the required precision tightens. Meanwhile, the 2-qubit CNOT gate is already a member of the Clifford group and requires *zero* T gates exactly. The asymmetry is already visible at the physical level.

Layer 2: Gate injection. The situation worsens once we impose a realistic constraint: T gates cannot always be applied directly to the qubit of interest. In many architectures, non-Clifford gates must be *injected* via ancilla qubits and gate teleportation circuits [4]. This introduces new overheads—ancilla count, 2-qubit gate depth, and probabilistic repetition (repeat-until-success protocols)—none of which afflict the CNOT gate, which remains directly available. What was a cost difference in T-count has now become a structural overhead difference in circuit architecture.

Layer 3: Logical qubits and fault tolerance. The deepest layer of the argument emerges when we move from a physical qubit to a *logical* qubit protected by quantum error correction. The surface code, one of the leading candidates for fault-tolerant quantum computing, has a natural set of transversal gates—operations that can be applied qubit-by-qubit across the code block without spreading errors. The CNOT gate is transversal in the surface code. Small-angle Z-rotations are not. Implementing $R_z(\pi/2^k)$ on a logical qubit requires a full state injection protocol, post-selection, and repeated syndrome extraction rounds, as studied in the STAR architecture [**akahoshi˙star˙2023**, 3, 7]. The resource overhead is substantial, and it grows with the required precision of the rotation—a cost that a logical CNOT simply does not pay.

Taken together, these three layers give a precise and satisfying answer to the ghost's claim. The 1-qubit gate $R_z(\pi/2^k)$ is harder not because it is intrinsically complex, but because the infrastructure of fault-tolerant quantum computing is built around gates that respect the symmetries of error-correcting codes—and small arbitrary rotations do not. The 2-qubit CNOT, by contrast, fits naturally into that infrastructure at every level.

The remainder of this write-up is organized as follows. Section **??** develops the synthesis story for physical qubits. Section **??** addresses gate injection and the overhead of non-Clifford gates without direct access. Section **??** lifts the argument to the logical qubit level using the Steane code. Section **??** presents our simulations of the STAR architecture and the syndrome extraction circuits that underpin it. Section **??** draws the threads together.

2 Synthesis Challenges

2.1 Learn the language of Clifford+T

The Clifford gates preserve structure in the sense that they map a stabilizer state onto another stabilizer state, making them classically possible to simulate. The T gate breaks this

symmetry by introducing a phase outside the stabilizer formalism, which makes it difficult for classical computers to simulate. Consider the following 1-qubit and 2-qubit circuit Clifford + T simulations using as examples.

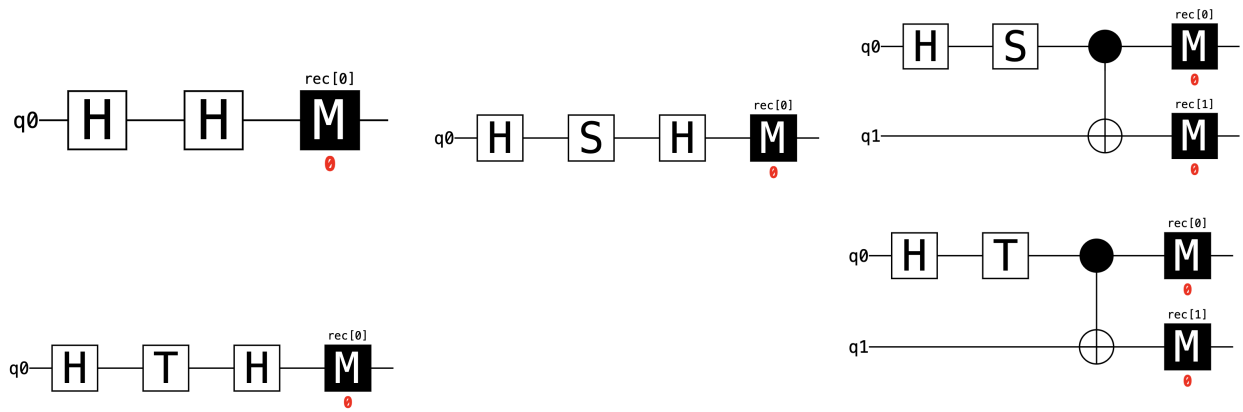

Figure 1: 1 and 2 qubit circuit simulations to describe Clifford (H, S, CNOT) and non-Clifford (T) gates

Figure 2: Statevector visualizations for HSH and HTH circuit simulations

The top circuits of 1 are all made up of Clifford gates, which basically maps the initialized qubit in the Pauli eigenstates to other Pauli eigenstates. However, the bottom circuits leads to a transformation outside this stabilizer formalism, leading to what is konwn as the magic state for the two-qubit simulation.

2.2 Synthesize the rotation family

This challenge focused on implementing Z-rotations

$$R_z(\pi/2^n), n \in \{0, 1, 2, 3, 4, 5\} \tag{2}$$

using only H, S, and T gates on a single qubit. For $n \in \{0, 1, 2\}$, the solutions are exact and trivial. Namely, $R_z(\pi/2^0) = R_z(\pi)$ is simply the Pauli Z gate [1] which is equivalent to S^2, $R_z(\pi/2^1) = R_z(\pi/2)$, is equivalent to the S gate, and $R_z(\pi/2^2) = R_z(\pi/4)$ is equivalent to the T gate.

To perform the approximation, we used the `gridsynth` algorithm [5, 6], which is provably optimal in T-count for a given precision ε. The algorithm takes a target rotation angle and

[1] up to a global phase

returns a gate string over the alphabet $\{H, S, T, X, W\}$, where W corresponds to a global phase of $e^{i\pi/4}$. These gate strings are applied to the circuit in reverse order, as `gridsynth` outputs the operator product from left to right. Given that we cannot use X gates, we convert each of the X gates that `gridsynth` returns to the sequence $HSSH$ using the identities $HZH = X$ and $Z = S^2$.

A subtle but important detail arises with global phases. By default, `gridsynth` approximates the z-rotation on the nose, meaning it targets the exact matrix which is not necessarily the same as the standard gate definitions up to global phase. For example, with $\theta = \pi/4$, the default output does not return a single T gate, but instead will give an approximation of

$$R_z(\pi/4) = \begin{pmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{pmatrix} \tag{3}$$

which differs by a global phase from T. Without accounting for this, the algorithm unnecessarily spends additional gates to match the exact diagonal entries rather than recognizing that a single T suffices. To resolve this, we invoked `gridsynth` with the `--phase` flag, which instructs the algorithm to approximate the rotation up to a global phase. With this option, the $n = 2$ case correctly returns a single T gate, and all other cases benefit from shorter sequences since the algorithm has the freedom to choose the most efficient global phase convention.

To assess the quality of each synthesized approximation, we used the global-phase-invariant distance

$$d(U, V) = \sqrt{1 - \frac{|\mathrm{Tr}(U^\dagger V)|}{2}}, \tag{4}$$

where U is the target unitary and V is the synthesized approximation. This metric is zero when U and V agree up to global phase, and increases as the approximation worsens.

We benchmarked the synthesis over an extended range $n \in \{0, 1, \ldots, 33\}$ to observe how the approximation quality scales as the target angle becomes exponentially smaller. The results are shown in Figure 3 and confirm that the approximations for $n \in \{0, 1, 2\}$ are exact.

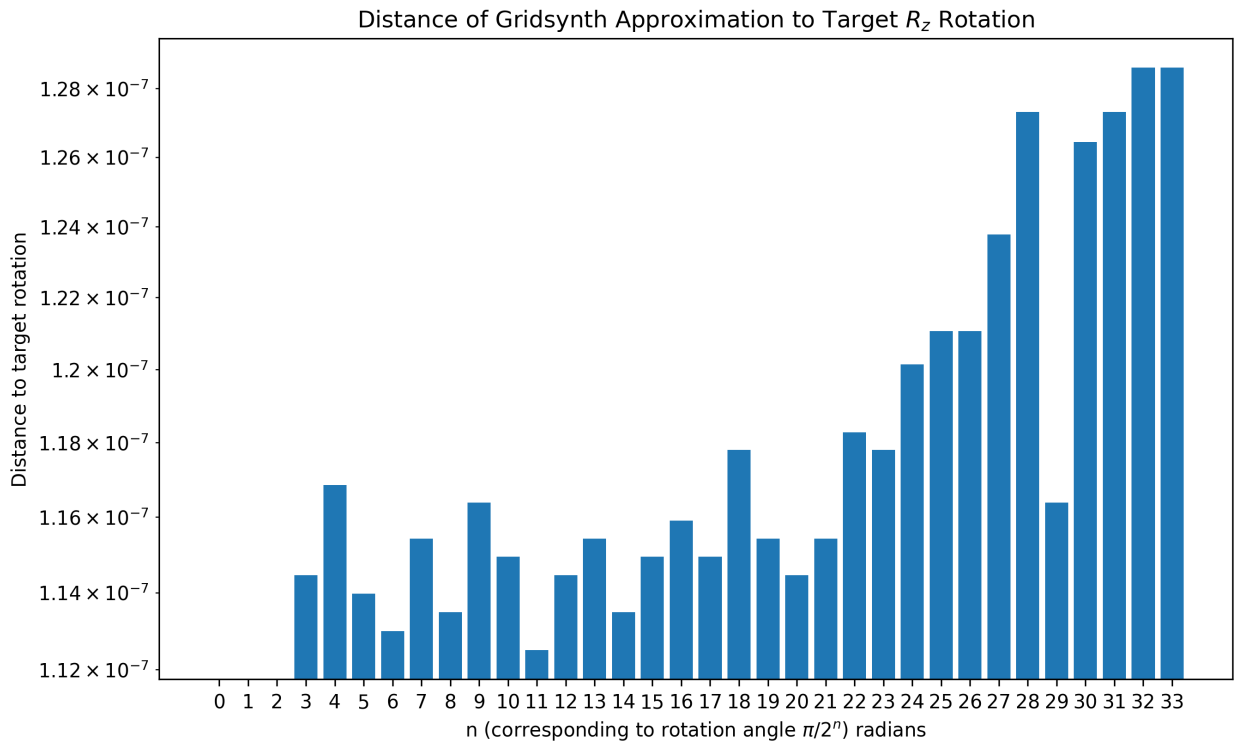

Figure 3: Global-phase-invariant distance $d(U, V)$ between the target rotation $R_z(\pi/2^n)$ and its `gridsynth` approximation, plotted on a logarithmic scale. For $n \leq 2$, the synthesis is exact ($d = 0$, not shown on the log scale). For $n \geq 3$, the distance remains small and roughly uniform, reflecting the default precision threshold of `gridsynth`.

Several observations emerge from this analysis. First, the cases $n \leq 2$ are synthesized exactly with T-count zero (since Z, S, and T are themselves Clifford or the single non-Clifford generator). Second, for $n \geq 3$, the `gridsynth` algorithm produces approximations whose distance hovers near its default precision floor; the algorithm is not struggling to approximate these rotations, but it is spending T gates to do so. The Solovay Kitaev theorem guarantees that any single-qubit unitary can be approximated to precision ε using $O(\log^c(1/\varepsilon))$ gates from a universal set, but `gridsynth` achieves the tighter scaling of $4\log_2(1/\epsilon) + O(\log(\log(1/\varepsilon)))$ in T-count in the typical case [6].
The central lesson of this section is that while any single-qubit rotation can in principle be approximated arbitrarily well using Clifford+T, the non-Clifford resource cost (measured in T-count) is nonzero and grows with the desired precision. This cost is modest for a single physical qubit, but as we will see in Parts 3 and 4, the situation becomes dramatically worse when T gates cannot be applied directly or when the qubit is encoded in an error-correcting code.

2.3 Non-Clifford gates are expensive

In Part 2, T gates were applied directly to the data qubit. In practice, however, hardware and error-correction constraints often prevent direct application of non-Clifford gates to the

data qubit. This section explores how to recover the effect of a T gate using only Clifford operations on the data qubit, with the help of an ancilla qubit on which T gates are permitted. The protocol used is gate teleportation. For each T gate in the gridsynth sequence, the injection circuit proceeds as follows. First, the ancilla is prepared in the magic state $T|+\rangle$ by applying H then T. Next, a CNOT is applied with the data qubit as control and the ancilla as target. The ancilla is then measured. If the measurement outcome is 1, a corrective $S^\dagger$ gate is applied to the data qubit (a feed-forward operation), and the ancilla is reset via an X gate so it can be reused for subsequent T injections. All other gates in the `gridsynth` sequence (H, S, and X) are applied directly to the data qubit as before, since these are Clifford operations.

This scheme uses a single ancilla qubit throughout, resetting it after each injection rather than allocating a fresh ancilla per T gate. Each T gate in the original sequence contributes one CNOT (a 2-qubit gate), one mid-circuit measurement, and potentially one feed-forward correction to the resource cost of the injection circuit. The Clifford gates H, S, and X remain free of 2-qubit overhead.

The circuits were constructed using Bloqade Squin and simulated with PyQrack via the StackMemorySimulator. A key detail in the simulation is that $R_z(\theta)|0\rangle = e^{-i\theta/2}|0\rangle$, which differs from $|0\rangle$ only by a global phase. Since global phases are unobservable, applying the rotation family directly to $|0\rangle$ produces a state indistinguishable from the input regardless of whether the synthesis is correct. To make the effect of the rotation observable, an H gate was prepended to each circuit, placing the qubit in the $|+\rangle$ state before the synthesized rotation is applied. In this basis, $R_z(\theta)$ produces a relative phase between the $|0\rangle$ and $|1\rangle$ components that is visible in the density matrix.

To validate the injection protocol, both the direct-application circuit (with T gates on the data qubit) and the injection circuit (with T gates only on the ancilla) were simulated for $n \in \{0, 1, \ldots, 9\}$. The reduced density matrix of the data qubit was extracted from each simulation and compared. The two density matrices agreed to within 10^{-5} for all values of n tested, confirming that the injection protocol faithfully reproduces the target rotation.

The infidelity $1 - \mathrm{Tr}(\rho_{\mathrm{direct}}\rho_{\mathrm{inject}})$ between the direct and injected density matrices was also computed and is shown in Figure 4. The values remain at the 10^{-5} level across all n, consistent with the finite precision of the `gridsynth` approximation rather than any error introduced by the injection protocol itself.

The resource overhead introduced by injection is summarized in Table 1. For a `gridsynth` sequence targeting $R_z(\pi/2^3)$ with default precision, the injection circuit requires on the order of tens of CNOTs and feed-forward corrections, whereas the direct circuit requires none. The circuit depth also increases due to the ancilla preparation, entangling gate, measurement, and conditional correction steps that accompany each injected T gate. This overhead is the price of restricting non-Clifford operations to the ancilla, and it grows linearly with the T-count of the synthesized sequence.

The central observation is that even though the injection protocol introduces no additional approximation error beyond what `gridsynth` already contributes, it converts each T gate from a single 1-qubit operation into a multi-step procedure involving 2-qubit entanglement, measurement, and classical feed-forward. A rotation that was trivial to implement with direct T access becomes substantially more expensive when that access is removed.

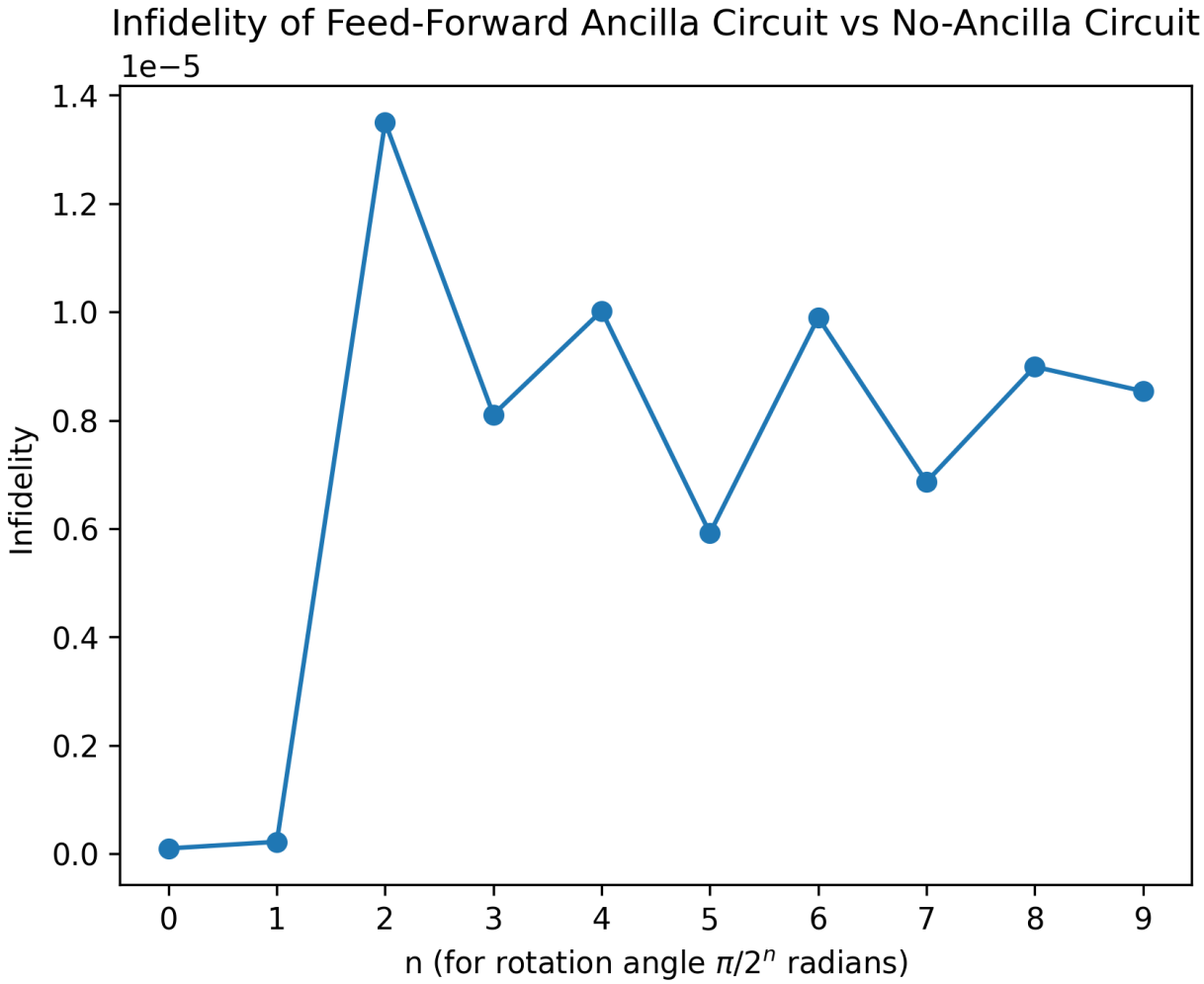

Figure 4: Infidelity between the density matrices produced by direct T-gate application and T-gate injection via ancilla, for target rotations $R_z(\pi/2^n)$ with $n = 0, \ldots, 9$. The small residual infidelity (order 10^{-5}) reflects `gridsynth` approximation error, not injection error.

Table 1: Resource comparison between direct application and injection for a representative `gridsynth` sequence. The ancilla count is 1 (constant) since the ancilla is reused.

Resource	Direct	Injection
Ancilla count	0	1
2-qubit gate count	0	$= T$-count
Feed-forward operations	0	$\leq T$-count
Circuit depth increase	0	$\leq 2 \times T$-count

2.4 Move from one physical qubit to one logical qubit

With the T gate teleportation developed in the previous section, there is now a potential framework for injecting non-clifford gates onto qubits which be restricted in operations. This becomes very important within the regime of logical qubits. This section explores how the [[7,1,3]] Steane code can be implemented to create single error correcting logical qubits, and how we can do gate operations on these logical qubits.

To start, the Steane code is a stabilizer code with 7 physical qubits, with the ability to correct 1 error, with a distance of 3. The Steane code implemented for our exploration allows for an arbitrary state to be inputted on qubit 0 which gets encoded among the other 6 physical qubits. The encoding circuit utilized which enables this property can be seen in Figure 5, implemented in Quirk Circuit Simulator. This encoding ciruit utilizes a combination of CNOTs and Hadamard Gates to entangle the other 6 physical qubits with each other and the arbitrary state qubit. The Hadamard gates seem to allow for entanglement via phase kickback effects.

With this encoding circuit we now have a logical qubit which can be corrected for one error utilizing a syndrome extracting circuit and a classically controlled decoding circuit which applies corrections based off the decoded syndromes. The Steane code utilizes 6 generators of the syndromes, 3 X-Type and 3 Z-Type generators. This section does not cover much of the actual error correcting portion of this code, as for our purposes, it is more important that we discover how to manipulate these entangled physical qubits as one encoded logical qubit.

To understand how to manipulate the logical qubit, it is essential understand the properties of a transversal gate set of a error correcting code. The transversal gate set is a set of gates which can be applied across all of the physical qubits of the logical qubit which directly applies the effect of the gate on the logical qubit as a whole. For the steane code this set is includes all Clifford Group gates. This means that for the universal gate set available, H, S, T, CX, the T gate is special in that it cannot be applied transversally to the logical qubit and must be teleported in, from an ancilla logical qubit to the main logical qubit. All other gates in the utilized universal gate set are within the Clifford group and can thus be applied transversally.

To apply the special non-clifford T gate, an approach very similar to the T gate teleportation from the previous section was tested, but with physical qubits replaced with logical qubits. In order to load the ancilla logical qubit with a T state, the special arbitrary state encoding property of the implemented Steane encoding circuit allows one to load a T state on a physical qubit and then encode that T state into a logical qubit. With this ancilla circuit, one can apply a very similar framework from the previous section, a CNOT gate is applied across all physical qubits of the two logical qubits with the ancilla being the target and the main logical qubit as the control. Then, all qubits of the ancilla are measured, where a feed-forward of that measurement is applied on a per physical qubit basis, where individually S gates are classically implemented if a 1 is returned for the individual physical ancilla qubit. This implementation of the T gate teleportation for logical qubits was compared against a T state encoded onto a logical qubit and was found to match within an order of magnitude of 10^-5.

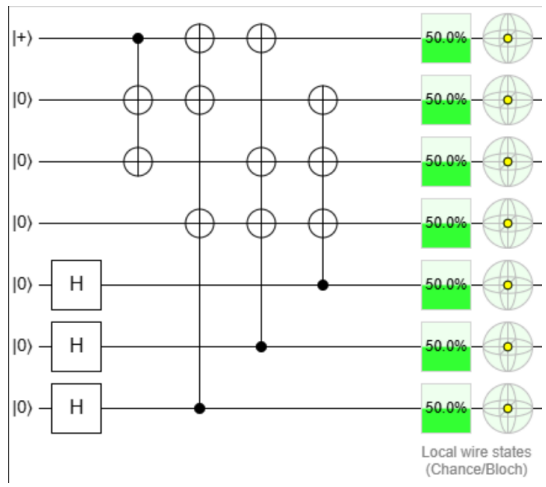

Figure 5: [[7,1,3]] Stean-code for single logical qubit encoding

3 STAR Challenges

3.1 Get comfortable using Tsim

This part focused on building and simulating a distance-3 surface code using `Tsim`, QuEra's Clifford circuit simulator, with the goal of understanding stabilizer structure and syndrome extraction.

Code Layout

The distance-3 surface code encodes one logical qubit into nine physical data qubits arranged on a 3×3 grid, labeled $D_0, \ldots, D_8$. Error detection is performed by four Z-type and four X-type stabilizer operators, each acting on a 2×2 block of neighboring data qubits:

$$\begin{aligned} Z_0 &= Z_{D_0} Z_{D_1} Z_{D_3} Z_{D_4}, & Z_1 &= Z_{D_1} Z_{D_2} Z_{D_4} Z_{D_5}, \\ Z_2 &= Z_{D_3} Z_{D_4} Z_{D_6} Z_{D_7}, & Z_3 &= Z_{D_4} Z_{D_5} Z_{D_7} Z_{D_8}, \end{aligned} \tag{5}$$

and analogously for the X-type stabilizers $X_0, \ldots, X_3$ acting on the same four data-qubit subsets.

Single-Stabilizer Building Blocks

Each Z stabilizer is measured by entangling an ancilla qubit a, initialized in $|0\rangle$, with the four data qubits via CNOT gates with the data qubits as controls:

$$\mathrm{CNOT}_{D_i,a} \quad \text{for each } D_i \in \mathrm{support}(Z_k). \tag{6}$$

Measuring a in the Z basis then yields the parity $\langle Z_{D_i} \rangle$ of the stabilizer. An X error on any data qubit in the support anti-commutes with the stabilizer and flips the measurement outcome from $+1$ to -1, confirming that Z-type checks detect X-type errors.

X stabilizers are measured analogously, with the ancilla prepared in $|+\rangle$ via a Hadamard and the CNOT directions reversed (ancilla as control, data as target), followed by a final Hadamard on the ancilla before measurement:

$$H_a \to \mathrm{CNOT}_{a,D_i} \to H_a \to M_a. \tag{7}$$

This implements a measurement of the X-type parity. A Z error on any data qubit in the support anti-commutes with the stabilizer and again flips the outcome, confirming that X-type checks detect Z-type errors.

Full Syndrome Rounds

All four Z stabilizers and all four X stabilizers were then assembled into full syndrome rounds over the complete 3×3 layout, using 9 data qubits and 4 ancilla qubits per round (13 total). The syndrome of a single round is a 4-bit binary string $s = (s_0, s_1, s_2, s_3)$, where $s_k = 0$ indicates the stabilizer Z_k (or X_k) is satisfied and $s_k = 1$ indicates a violation.
We verified the locality of the syndrome pattern under injected errors. An X error on the center qubit D_4 is in the support of all four Z stabilizers and produces the all-ones syndrome $s = (1, 1, 1, 1)$, while an X error on the off-center qubit D_1 is shared only between Z_0 and Z_1, yielding $s = (1, 1, 0, 0)$.

Two-Round Syndrome Extraction and Detection Events

To detect when an error *occurs*, rather than merely where the code is in an error state, we implemented two sequential rounds of syndrome extraction. Each round uses its own set of four fresh ancilla qubits,[2] so that rounds 1 and 2 produce measurement vectors $s^{(1)}$ and $s^{(2)} \in \{0, 1\}^4$ respectively.
A *detection event* is defined as a change in a stabilizer's measurement outcome between consecutive rounds:

$$e_k = s_k^{(1)} \oplus s_k^{(2)}, \tag{8}$$

where $\oplus$ denotes addition modulo 2. In the absence of errors, $s^{(1)} = s^{(2)}$, so $e = 0$. An error injected *between* rounds leaves $s^{(1)}$ unaffected but flips the relevant bits in $s^{(2)}$, producing a nonzero event vector e. We verified this: inserting an X error on D_4 between rounds yields the 8-bit combined detection event vector

$$e = (e^Z \,|\, e^X) = (1\,1\,1\,1 \,|\, 0\,0\,0\,0), \tag{9}$$

where the four Z-stabilizer events are all triggered and the X-stabilizer events remain zero, consistent with the fact that an X error commutes with all X stabilizers.
For clarity, Z-type and X-type syndrome extraction were implemented and validated separately. The final 8-bit combined detection event vector was assembled from the two independently validated extractors.

[2] In a real repeated-extraction circuit, ancilla qubits would be reset and reused between rounds. Here, fresh ancillas are allocated per round to avoid implementing mid-circuit reset, using 17 qubits total instead of 13.

3.2 Estimate STAR Fidelities

To estimate the fidelity of the STAR architecture under noise, we simulated the pre-built STAR circuits provided in the challenge assets using Tsim. The circuits are parameterized by the code distance $d \in \{3, 5, 7\}$ and a logical rotation angle. Each circuit file contains a Stim-format template with placeholders for the physical and logical rotation angles.

The relationship between the logical rotation angle θ_L (in units of π) and the physical rotation angle applied to each of the d physical qubits is nonlinear. It is given by

$$\theta_{\text{phys}} = \pm \frac{2}{\pi} \arctan\left(\tan\left(\frac{\theta_L \pi}{2} \right)^{1/d} \right), \tag{10}$$

where the sign depends on $d \bmod 4$. This formula distributes the logical rotation across d physical rotations such that their combined effect on the code block produces the desired logical angle.

For each combination of distance and logical angle, the simulation proceeded as follows. The physical angle was computed and substituted into the circuit template. The circuit was then compiled into a detector sampler using Tsim. A detector error model was extracted from the circuit and used to construct a minimum-weight perfect matching (MWPM) decoder via PyMatching. One million shots were sampled per configuration.

To isolate the logical error rate of the rotation gadget from syndrome extraction errors, we post-selected on shots where all stabilizer detectors in the first $3(d^2 - 1)$ detector slots reported no detection events. This filters to cases where the stabilizer measurements were consistent, so that any remaining logical error can be attributed to the rotation itself rather than to faulty syndrome extraction. Among the post-selected shots, the logical error rate was computed as the fraction of shots where the decoded observable disagreed with the sampled observable. This rate was then normalized by the code distance d to facilitate comparison across distances.

The simulation was run in a continuous loop over logical angles $\theta_L = 0.8/2^x$ for $x \in \{0, 1, \ldots, 11\}$, spanning from 0.8π down to approximately $4 \times 10^{-4}\pi$. For small rotation angles and for the largest code distances, the very low logical error rates necessitated a correspondingly high number of shots before errors were detected. Results were thus written incrementally to a CSV file, allowing us to adjust the required shot count to balance precision with compute time. The resulting logical error rate per distance as a function of logical rotation angle is shown in Figure 6.

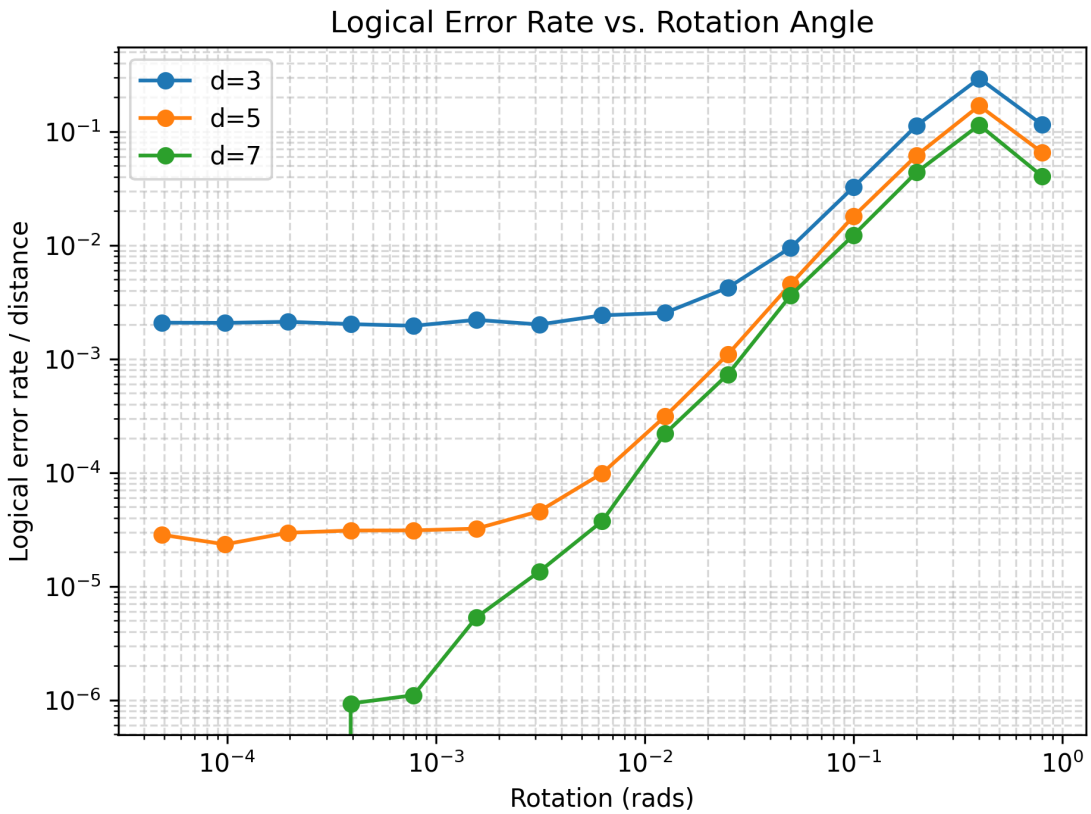

Figure 6: Logical error rate per code distance as a function of logical rotation angle for STAR circuits at distances $d = 3, 5, 7$. Both axes are logarithmic. Smaller rotation angles generally yield lower logical error rates, and increasing the code distance suppresses errors, demonstrating that the STAR architecture achieves error suppression that improves with distance. The logical error rate drops to zero for $d = 7$ at the lowest rotation angles, indicating that not enough shots were sampled to detect an error.

The results reproduce the qualitative behavior of the reference fidelity plot provided in the challenge. At large rotation angles, the logical error rate is highest and roughly comparable across distances. As the rotation angle decreases, the error rate drops, and higher-distance codes achieve progressively lower error rates. This confirms that the STAR transversal rotation gadget benefits from increasing code distance in the expected way: the error suppression scales with d, consistent with the threshold behavior of the surface code.

3.3 Teleport a non-Clifford rotation into a logical qubit

The key asymmetry of the STAR architecture is the distinction between Clifford and non-Clifford operations at the logical level. Clifford gates (such as H, S, and CNOT) are transversal in the surface code: applying the gate to every physical qubit in the code block implements the corresponding logical gate. Non-Clifford gates like $R_z(\theta)$ for arbitrary θ cannot be applied this way. Instead, the rotation must be teleported into the logical qubit using an

ancillary logical block.

The teleportation protocol works as follows [2]. Two distance-d surface code blocks are used: a main block encoding the logical state $|\psi\rangle_L$, and an ancilla block. Both blocks are initialized in the $|+\rangle$ state by applying transversal R_X to all qubits. The non-Clifford rotation $R_z(\theta_{\text{phys}})$ is then applied to the physical qubits in the left column of the ancilla block (the support of the logical Z operator). A transversal CNOT from the main block to the ancilla block entangles the two. The ancilla data qubits are then measured in the Z basis, and the parity of these measurements determines whether the teleportation succeeded. Shots where this parity is nonzero are discarded via post-selection, since Tsim does not support feed-forward operations. For the post-selected shots, the logical X observable is read out via a multi-qubit Pauli product measurement on the top row of the main block.

For the $d = 3$ surface code, this circuit uses 34 qubits total: 17 per block (9 data qubits and 8 syndrome ancillas). The physical rotation angle per qubit is $\theta_{\text{phys}} = \theta_L/d$, distributing the logical angle across the d physical qubits on the Z_L support.

To verify that the teleportation protocol correctly implements the intended logical rotation, we compared the STAR logical qubit against a single physical qubit undergoing the same sequence of operations. Since the simulation is noiseless, a single physical qubit is itself a trivial "logical qubit," so any discrepancy between the two would indicate an error in the circuit construction rather than a noise effect.

For both the STAR circuit and the reference physical qubit, we sampled the probability of measuring the $+X$ and $-X$ eigenstates. These probabilities define a partial characterization of the state. The overlap between the STAR logical state and the reference state was computed from these X-basis probabilities. Ideally, one would repeat this measurement in the Y and Z bases as well to fully reconstruct the state vector via tomography. Due to time constraints, only the X-basis measurement was completed.

Figure 7 shows $|\langle\psi_{\text{ref}}|\psi_{\text{STAR}}\rangle|^2$ as a function of the logical rotation angle, swept from 0.001π to π. The overlap remains very close to 1 across the full range, confirming that the teleportation protocol faithfully implements the target R_z rotation at the logical level. The post-selection rate was approximately 50%, consistent with the expected behavior of the protocol in the noiseless setting.

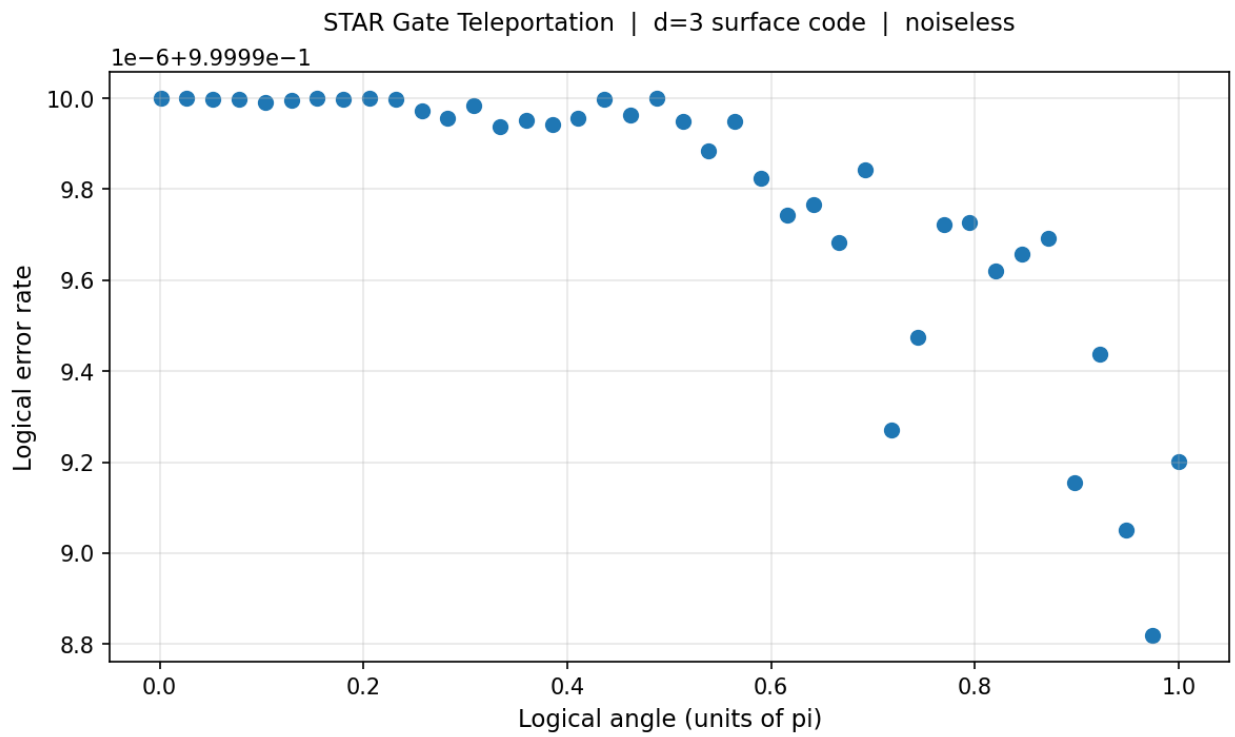

Figure 7: Squared overlap between the STAR logical qubit state and a reference physical qubit state as a function of logical rotation angle (units of π), for a $d = 3$ surface code in the noiseless setting. The overlap remains near 1 across the full range, verifying the correctness of the teleportation protocol. The small nonzero logical error rate visible in the accompanying error rate plot grows with the rotation angle, perhaps due to finite post-selection statistics.

3.4 Apply STAR in a Larger Algorithm

This part applied the teleported non-Clifford rotation framework to a concrete algorithm: a 4-qubit Quantum Fourier Transform (QFT4), where all non-Clifford phase gates are supplied via postselected teleportation gadgets rather than applied directly.

QFT4 Non-Clifford Content

The 4-qubit QFT is decomposed into Hadamard gates and controlled phase gates CR_k, which apply a phase of $e^{2\pi i/2^k}$ to the target qubit conditioned on the control. For $n = 4$ qubits, three distinct controlled phases appear:

$$\begin{aligned} CR_2 &: \theta = \pi/2 \quad \text{(Clifford, direct)}, \\ CR_3 &: \theta = \pi/4 \quad \text{(non-Clifford, teleported)}, \\ CR_4 &: \theta = \pi/8 \quad \text{(non-Clifford, teleported)}. \end{aligned} \tag{11}$$

CR_2 is equivalent to a controlled-S gate and lies within the Clifford group, so it is applied directly. CR_3 appears twice (qubits $q_0 \to q_2$ and $q_1 \to q_3$) and CR_4 once ($q_0 \to q_3$), giving a total of **3 teleported non-Clifford rotations** per complete QFT4 circuit.

Gate Decomposition

Each controlled phase is implemented via the standard Clifford-plus-R_z decomposition:

$$CR(\theta) \;=\; \mathrm{CNOT}_{c,t} \;\cdot\; R_z(\theta)_t \;\cdot\; \mathrm{CNOT}_{c,t}, \tag{12}$$

where c is the control qubit and t is the target. The two CNOT layers are Clifford and applied directly. For non-Clifford angles ($\theta = \pi/4$ and $\pi/8$), the $R_z(\theta)$ on the target is supplied by the postselected teleportation gadget described in Star 3. If the gadget rejects, the full circuit run is discarded and retried from scratch.

Postselection Overhead

Each teleportation gadget independently accepts with probability 1/2, since postselection keeps only the branch requiring no Pauli correction. With 3 independent non-Clifford rotations, the per-shot acceptance probability is:

$$p_{\text{accept}} \;=\; \left(\frac{1}{2}\right)^3 \;=\; 0.125. \tag{13}$$

We confirmed this empirically over 5000 trials, finding an acceptance rate of 0.127, in close agreement with theory. Correctness was verified independently: for all six computational basis inputs $\{|0000\rangle, |0001\rangle, |0010\rangle, |0100\rangle, |1000\rangle, |1111\rangle\}$, accepted shots matched the direct reference QFT4 output up to global phase.

Scaling

The acceptance cost generalises immediately. For a circuit containing k teleported non-Clifford rotations:

$$p_{\text{accept}} \;=\; \left(\frac{1}{2}\right)^k. \tag{14}$$

To measure this directly, we ran k independent postselected teleportations on a single qubit for $k = 1, \ldots, 12$, with 2000 trials per point. Figure 8 shows the theoretical and empirical acceptance probability. The two curves agree closely for $k \leq 7$; at larger k the empirical estimates become noisy because the expected number of accepted shots falls below ten and statistical fluctuations dominate. The QFT4 case ($k = 3$) is marked for reference. The implication for larger algorithms is immediate: a general n-qubit QFT requires $O(n^2)$ non-Clifford controlled phases, making naive postselection completely impractical at scale.

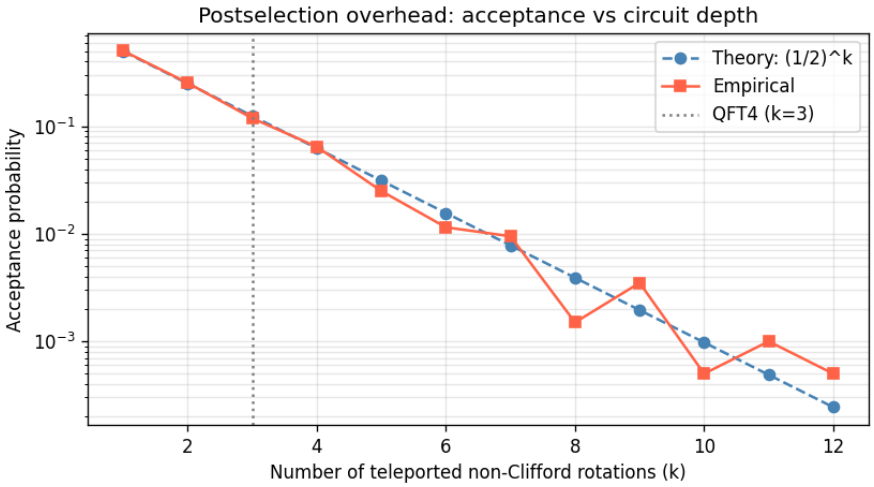

Figure 8: Acceptance probability vs. number of teleported non-Clifford rotations k, on a log scale. Theory $(1/2)^k$ (blue dashed) and empirical estimate from 2000 trials per point (red solid) agree well for $k \leq 7$. The vertical dotted line marks $k = 3$ (QFT4).

For a general n-qubit QFT, the number of non-Clifford controlled phases grows as $O(n^2)$, making the postselected acceptance rate doubly exponentially small in n. This confirms that postselection alone cannot scale: practical implementations require feed-forward corrections, repeat-until-success protocols, or the STAR resource-state injection framework to avoid the exponential overhead.

4 AI Usage

We used AI to help research the background necessary to understand the challenges, draft and debug code, and draft this paper.

5 References

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